pmsimstats-ng Documentation Index

pmsimstats team

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This document indexes the technical documentation for the `pmsimstats-ng` simulation framework. Documents are grouped by their role in two narratives: (a) the Hendrickson et al. (2020) research program on biomarker-moderated treatment response in aggregated N-of-1 trial designs, and (b) the development path from the original codebase to the current dual-architecture framework.

1 Research narrative

These documents describe the scientific goals, statistical methods, and simulation results for the Hendrickson et al. (2020) power analysis of predictive biomarker-treatment interactions.

Document	Scope
`04`	Revised power analysis with corrected covariance structure and carryover model (Architecture B). Includes commit-by-commit comparison appendix.
`06`	AR(1) residual correlation: transition from `lmer` to `nlme` with `corCAR1`. Applies to both architectures.
`07`	Positive definiteness failures in the MVN covariance matrix (Architecture B specific).
`08`	Derivation of the λ_{cor} BM-BR correlation decay during off-drug periods (Architecture B specific).
`09`	Analysis of the step-function BM-BR correlation artifact in the original code (Architecture B specific).
`18`	Response parameter sensitivity (Figure 5): original vs. revised implementations (Architecture B).

2 Dual-architecture framework

These documents describe the two data-generating process architectures for the biomarker-treatment interaction and the implementation of Architecture A (direct mean moderation).

Document	Scope
02	Two DGP architectures for biomarker-treatment interactions: theoretical comparison, empirical results, biological assumptions, and analysis strategies.
19	Architecture A implementation: four attempts, correct calibration, PD constraint removal, and Hendrickson publication parameter evaluation.
21	Architecture comparison summary: brief empirical power contrast and biological assumptions for each architecture. Companion summary to doc 02.

3 Codebase reference

Document	Scope
01	Codebase overview: repository structure, R package functions, analysis scripts, key parameters.
13	Figure 4 walkthrough: original publication methodology (*pre-audit historical reference*).

4 Three implementation collections

The `implementations/` directory contains three parallel, complete implementations of the simulation framework, each supporting different architectures and coding styles:

Collection	Scope
`original`	Architecture B (MVN) only; corrected Hendrickson publication code; data.table style. Baseline reference implementation.
`original-extended`	Architecture A + B (both); original code extended with `dgp_architecture` parameter; data.table style. Production use.
`tidyverse`	Architecture A + B (both); complete tidyverse reimplementation; tidyverse style. Modern alternative.

Each collection has a `README.md` with parameter documentation, usage examples, and validation notes. All three accept the same core parameters (`c.bm`, `carryover_t1half`, `lambda_cor`); only `original-extended` and `tidyverse` support the `dgp_architecture` parameter for Architecture A (mean moderation) vs. Architecture B (MVN).

5 Collaborator deliverables

Document	Scope
03	Audit and revision report: status summary for collaborator presentation.
10	Carryover analysis model assessment: standard methodology and application to pmsimstats.
14	Trial analysis implementation guide: response to collaborator goals and priorities.

6 Internal development

Document	Scope
15	Tidyverse alignment plan: 9-step roadmap for matching the 2025 pipeline to the orig gold standard.
16	Tidyverse code review: dead code, duplication, and simplification opportunities.
17	Simplification plan: options for reducing complexity in the simulation framework.

7 Reference materials

Document	Scope
20	Annotated bibliography of N-of-1 simulation methodology literature; supports the white paper (02) and audit report (03).
22	Journal target recommendation for the dual-architecture white paper (02); publication strategy and venue analysis.

8 Executable reports

The Gompertz clinician guide (`analysis/report/11-gompertz-clinician-guide.Rmd`) contains executable R code and is rendered via `rmarkdown`. It resides in the zzc report directory rather than in `docs/`.

9 Removed documents

Document	Reason
05	Duplicate of 04 (removed).
11	Moved to `analysis/report/` (executable Rmd).
12	Number not assigned.

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Source: ~/prj/alz/10-pmsimstats-ng/pmsimstats-ng/docs/00-documentation-index.tex